Library Management System in Java

File Structure

```
LibraryManagementSystem/
│
├── Book.java
├── Library.java
└── Main.java
```

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1. Book.java

```java
public class Book {

    // Encapsulation
    private int bookId;
    private String title;
    private String author;
    private boolean isIssued;

    // Constructor
    public Book(int bookId, String title, String author) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.isIssued = false;
    }

    // Getters
    public int getBookId() {
        return bookId;
    }

    public String getTitle() {
        return title;
    }

    public String getAuthor() {
        return author;
    }
```

```java
    public boolean isIssued() {
        return isIssued;
    }

    // Issue book
    public void issueBook() {
        if (!isIssued) {
            isIssued = true;
            System.out.println("Book issued successfully.");
        } else {
            System.out.println("Book already issued.");
        }
    }

    // Return book
    public void returnBook() {
        if (isIssued) {
            isIssued = false;
            System.out.println("Book returned successfully.");
        } else {
            System.out.println("Book was not issued.");
        }
    }

    // Display Book Info
    public void displayBook() {
        System.out.println("--------------------------------");
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Author  : " + author);
        System.out.println("Status  : " + (isIssued ? "Issued" : "Available"));
    }
}
```

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2. Library.java

```java
import java.util.ArrayList;

public class Library {

    private ArrayList<Book> books = new ArrayList<>();

    // Add Book
    public void addBook(Book book) {
```

```
        books.add(book);
        System.out.println("Book added successfully.");
    }

    // Show All Books
    public void showBooks() {

        if (books.isEmpty()) {
            System.out.println("No books available.");
            return;
        }

        for (Book book : books) {
            book.displayBook();
        }
    }

    // Search Book
    public Book searchBook(int id) {

        for (Book book : books) {
            if (book.getBookId() == id) {
                return book;
            }
        }

        return null;
    }

    // Issue Book
    public void issueBook(int id) {

        Book book = searchBook(id);

        if (book != null) {
            book.issueBook();
        } else {
            System.out.println("Book not found.");
        }
    }

    // Return Book
    public void returnBook(int id) {

        Book book = searchBook(id);

        if (book != null) {
            book.returnBook();
```

```
        } else {
            System.out.println("Book not found.");
        }
    }
}
```

---

3. Main.java

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Library library = new Library();
        Scanner sc = new Scanner(System.in);

        int choice;

        do {

            System.out.println("\n===== LIBRARY MANAGEMENT SYSTEM =====");
            System.out.println("1. Add Book");
            System.out.println("2. Show Books");
            System.out.println("3. Issue Book");
            System.out.println("4. Return Book");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");

            choice = sc.nextInt();

            switch (choice) {

                case 1:

                    System.out.print("Enter Book ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();

                    System.out.print("Enter Book Title: ");
                    String title = sc.nextLine();

                    System.out.print("Enter Author Name: ");
                    String author = sc.nextLine();
```

```java
                Book book = new Book(id, title, author);

                library.addBook(book);
                break;

            case 2:

                library.showBooks();
                break;

            case 3:

                System.out.print("Enter Book ID to issue: ");
                int issueId = sc.nextInt();

                library.issueBook(issueId);
                break;

            case 4:

                System.out.print("Enter Book ID to return: ");
                int returnId = sc.nextInt();

                library.returnBook(returnId);
                break;

            case 5:

                System.out.println("Thank you for using the system.");
                break;

            default:

                System.out.println("Invalid choice.");
        }

    } while (choice != 5);

    sc.close();
  }
}
```

Compile and Run

Compile

```
javac *.java
```

Run

```
java Main
```

OOP Concepts Used

Concept	Usage
Class	Book, Library, Main
Object	Creating Book objects
Encapsulation	Private variables
Constructor	Book constructor
Methods	addBook(), issueBook()
ArrayList	Store books dynamically
Scanner	User input

Future Improvements

- Add MySQL database
- Add Login System
- Add GUI using Swing
- Add Student Records
- Add Fine Calculation
- Convert into Spring Boot Project